

Karanvir Singh

MERN Stack Developer — React.js — Next.js — Node.js — TypeScript — AI-Assisted Development
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Professional Summary

MERN Stack Developer with 4+ years of experience building scalable, production-grade web applications using React.js, Next.js, Node.js, Express.js, and MongoDB. Strong expertise in full-stack development, REST API design, authentication systems, role-based access control (RBAC), and performance optimization. Experienced in building admin dashboards, SaaS platforms, real-time applications, and modern responsive UI systems. Proficient in AI-assisted development, leveraging AI tools and LLMs to accelerate coding, debugging, testing, documentation, and solution development while maintaining code quality and engineering best practices.

Technical Skills

Frontend: React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Shadcn/ui, Material UI, Ant Design, Redux, Zustand, React Query, Vite

Backend: Node.js, Express.js, REST APIs, API Integration, Middleware, Server-Side Rendering

Database: MongoDB, MongoDB Atlas, Database Design, Query Optimization

Authentication & Security: JWT, OAuth 2.0, NextAuth, Role-Based Access Control (RBAC), Protected Routes

Realtime: Socket.io, WebSockets, Real-Time Communication

AI-Assisted Development: LLMs, AI Code Generation, Prompt Engineering, AI-Assisted Debugging, Code Refactoring, Test Generation, Technical Documentation, Cursor, Claude, Lovable, Replit, CodeRabbit

DevOps & Tools: Git, GitHub, GitHub Actions, Vercel, Render, Postman, Jira, CI/CD

Performance: Code Splitting, Lazy Loading, Memoization, API Optimization, Database Query Optimization, Core Web Vitals, SEO

Professional Experience

Software Engineer (MERN) – Lumino Guru

Nov 2024 – Present

- Engineered scalable React.js and Next.js applications using SSR, SSG, reusable component architecture, and API-driven workflows, improving application performance, SEO, and maintainability.
- Built reusable component libraries using Tailwind CSS and Shadcn/ui, improving development efficiency and UI consistency.
- Integrated REST APIs with optimized data handling, reducing frontend API latency and improving response stability.
- Implemented performance optimization techniques including code splitting, lazy loading, and memoization, improving Core Web Vitals scores.
- Collaborated with backend developers to design and consume scalable REST APIs for production applications.
- Participated in requirement analysis, technical design discussions, and sprint planning in Agile environment.
- Mentored junior developers and performed code reviews ensuring clean code practices and reusable architecture.
- Worked on debugging production issues and improving application stability across browsers and devices.

MERN Stack Developer – Wits Innovation Lab

May 2022 – Oct 2024

- Developed full-stack MERN applications using React.js, Node.js, Express.js, and MongoDB for enterprise clients.
- Designed and implemented RESTful APIs with proper validation, authentication, and error handling.
- Built secure authentication systems using JWT, OAuth, and Role-Based Access Control (RBAC).
- Developed responsive dashboards and admin panels with real-time data visualization.
- Improved application performance by optimizing API calls, database queries, and frontend rendering logic.
- Designed and integrated RESTful APIs with optimized data-fetching, caching, and error-handling strategies to improve frontend responsiveness and application reliability.
- Built reusable frontend components and custom hooks to reduce code duplication and improve maintainability.
- Worked in Agile teams using Jira and Git workflows, ensuring timely delivery of features.
- Participated in deployment pipelines using CI/CD tools on Vercel and Render.
- Fixed critical UI/UX bugs and reduced customer-reported frontend issues through systematic debugging.

Projects

JC Automax – Admin Portal (React.js, Node.js, MongoDB, Tailwind CSS, Shadcn)

- Built a complete admin portal for managing vehicles, users, and business partners.
- Implemented full CRUD operations for inventory and user management systems.
- Designed role-based access control (RBAC) for secure admin workflows.
- Developed analytics dashboard showing real-time insights on sales and inventory.
- Created responsive UI using Tailwind CSS and Shadcn for modern user experience.

JC Automax – Main Website (React.js, Tailwind CSS)

- Developed public-facing website showcasing car inventory with images and videos.
- Enabled admin-controlled content upload for dynamic listings.
- Built fully responsive UI optimized for mobile and desktop users.
- Improved user experience with clean and fast navigation flow.

HorseTelex Pedigree – Web Platform (Next.js, React.js, TypeScript, Tailwind CSS, Shadcn, Supabase)

- Built a production Next.js platform covering pedigree search, breeding tools, auctions, and sales/lease listings.
- Implemented secure API proxy layer and cookie-based auth so clients never expose upstream credentials.
- Developed complex horse pages with pedigree trees, offspring filtering/pagination, and virtual mating tools.
- Integrated i18n across 11 languages and a Supabase-backed sales section with mock-data fallbacks.
- Designed responsive admin/member UI using Tailwind CSS and Shadcn for consistent, modern workflows.

Postkodes – Logistics Platform (MERN Stack)

- Built scalable logistics platform for shipment tracking and management.
- Developed REST APIs for booking, tracking, and carrier operations.
- Created React dashboard for real-time logistics monitoring.

Flahyhealth – AI Healthcare Platform (Next.js + Node.js)

- Built AI-powered healthcare system with patient and admin modules.
- Implemented secure authentication and payment gateway integration.
- Used SSR/SSG for SEO optimization and performance improvement.

Community Schools – Real-Time Platform (MERN + Socket.io)

- Built real-time chat and collaboration platform using Socket.io.
- Developed backend APIs for posts, groups, and user interactions.
- Designed scalable architecture for concurrent users.

Encapsulate CRM Integration (Next.js + Node.js)

- Integrated Capsule CRM APIs for contact and account synchronization.
- Built secure authentication and protected routes.
- Developed modular architecture for scalable system design.

Education

MCA – Punjab Technical University	2018 – 2020
BCA – Panjab University	2015 – 2018